Practice Problems
Calculus-Based Physics II

Chapter 39 More on the Electric Potential

1) Assume that 10 J of work is needed to push an object with a net charge of negative 2 Coulombs, starting from rest at position A, along an electric field to position B. After the charged object is released, what is the kinetic energy of the object when it reaches its original starting position A?

$\Delta PE_E + \Delta KE = 0$

A) 20 J.
(B) 10 J.
C) zero.
D) 5 J.
E) impossible to predict without additional information.

2) A charged particle with charge + 5.0 μC is initially at rest. It is accelerated through a potential difference of 500 V. What is the kinetic energy of this charged particle?

$\Delta E = \Delta PE_E + \Delta KE = 0$

$\Delta PE = -q\Delta V$

$\Delta KE = KE_f - KE_i$ ($KE_i = 0$)

$-q\Delta V + KE_f = 0$

$KE_f = q\Delta V = 0.0025\ J$

3) An electron, initially at rest is accelerated through a potential difference of 550 V. What is the speed of the electron due to this potential difference? $m_e = 9.11 \times 10^{-31}$ kg

$\Delta KE + \Delta PE_E = 0$

$\Delta PE = -q\Delta V$

$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ ($v_i = 0$)

$\frac{1}{2}mv_f^2 + (-q\Delta V) = 0$

$v = \sqrt{\frac{2q\Delta V}{m}} = 1.39 \times 10^7\ m/s$

4) After a proton with an initial speed of 1.50×10^5 m/s has increased its speed by accelerating through a potential difference of 0.100 kV, what is its final speed? ($q = 1.60 \times 10^{-19}$ C , $m_{proton} = 1.67 \times 10^{-27}$ kg)

$v_i = 1.5 \times 10^5\ m/s$

$v_f = ?$

$\Delta V = 0.1 \times 10^3\ V$

$q = 1.6 \times 10^{-19}\ C$

$\Delta PE_E + \Delta KE = 0$

$\Delta PE = -q\Delta V$

$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

Solve For "v_f"

$\frac{1}{2}mv_f^2 = q\Delta V + \frac{1}{2}mv_i^2$

$v_f = \sqrt{v_i^2 + \frac{2q\Delta V}{m}}$

$v_f = 2.05 \times 10^5\ m/s$

5) The electron-volt is a unit of

A) electric field.
B) electric force.
(C) energy.
D) potential.
E) charge.

6) An object has a net positive charge of 22.22 nC, which best describes the potential created by this charged object at 4 meters from its center? (1 nC = 10^{-9} C)

A) 50 V
B) -50 V
C) + 50 V
D) + 12.5 V
E) -12.5 V

$V = \frac{Kq}{r} = +50V$ ↗ pos. charge

7) An object has a net negative charge of 22.22 nC, which best describes the potential created by this charged object at 4 meters from its center? (1 nC = 10^{-9} C)

A) 50 V
B) -50 V
C) + 50 V
D) + 12.5 V
E) -12.5 V

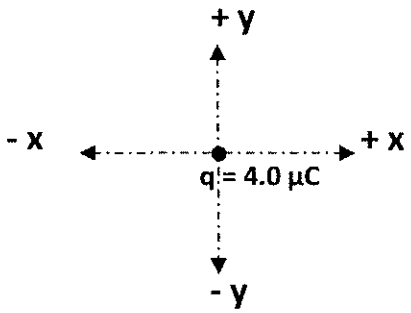
$V = \frac{Kq}{r} = -50V$ ↗ negative charge

8) A sphere with a net positive charge of 4.0 μC is situated at the origin of an *xy*-coordinate system. What is the potential difference between the points $x = 4.0$ m and $y = 2.0$ m because of this charge?

$\Delta V = V_2 - V_1 = \frac{Kq}{r_2} - \frac{Kq}{r_4}$

$\Delta V = (+18000V) - (+9000V)$

$\Delta V = +9{,}000V$

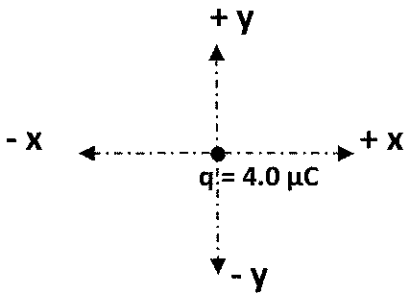


9) A sphere with a net negative charge of 4.0 μC p is situated at the origin of an *xy*-coordinate system. What is the potential difference between the points $x = 4.0$ m and $y = 2.0$ m because of this charge?

$\Delta V = V_2 - V_4 = \frac{Kq}{r_2} - \frac{Kq}{r_4}$

$\Delta V = (-18000V) - (-9000V)$

$\Delta V = -9000V$



10) A probe has a net positive charge q, if the potential difference between the positions $r_1 = 4$ m and $r_2 = 1$ m from the probe is 200 V, what is the charge on the probe?

$\Delta V = Kq\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$

$q = \frac{\Delta V}{\left[K\left(\frac{1}{r_2} - \frac{1}{r_1}\right)\right]} = 3 \times 10^{-8}\ C$

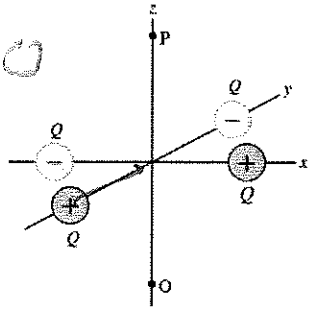
11) Four charged particles (two having a charge $+Q$ and two having a charge $-Q$) are arranged in the xy-plane, as shown in the figure. These particles are all equidistant from the origin. The electric potential at point P on the z-axis due to these particles, is (Assume same amount of charge Q but different types as shown.)

A) zero.
B) positive.
C) negative.
D) impossible to determine based on the information given.

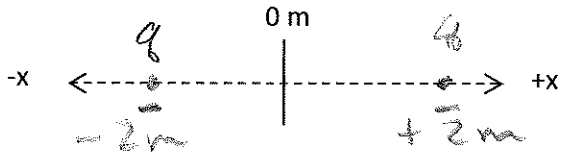
$V_T = V_- + V_- + V_+ + V_+ = 0$

$V_- = V_-$

$V_+ = V_+$



12) Two objects with a negative net charge of 4.0 μC are situated along the x-axis at $x_1 = -2.0$ m and $x_2 = 2.0$ m. What is the electric potential at the origin (0 m) of the xy-coordinate system?



$V_{Total} = V_{2m} + V_{-2m}$

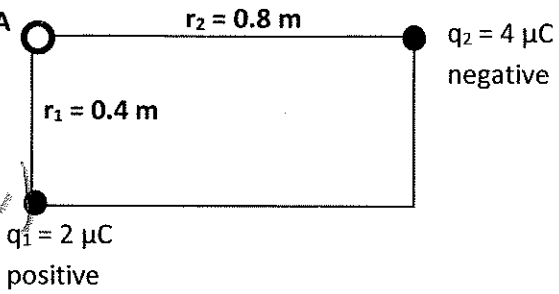
$V_{Total} = \frac{kq_1}{r_1} + \frac{kq_2}{r_2}$

$V_{Total} = -18000V + (-18000V)$

$V_{Total} = -36,000\,V$

13) Two point charges of magnitude 2.00 μC positive and 4.00 μC negative are placed at the opposite corners of a rectangle as shown in the figure. What is the potential at point A due to these charges?

$V_{Total} = V_A = V_1 + V_2 = \frac{kq_1}{r_1} + \frac{kq_2}{r_2}$

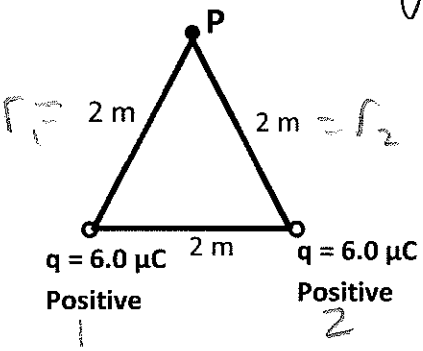


$V_A = (+45,000\,V) + (-45,000\,V)$

$V_A = 0\,V$

14) Two identical charged objects with a net charge of 6.0 μC are placed at the corners of the base of an equilateral triangle, as shown in the figure below, with the length of the sides being 2 m. What is the electric potential at the vertex, P, of the triangle due to these charges?

$V_P = V_1 + V_2 = \frac{kq_1}{r_1} + \frac{kq_2}{r_2}$



$r_1 = 2\text{ m} = r_2$

$V_P = +2.7 \times 10^4\,V + (+2.7 \times 10^4\,V)$

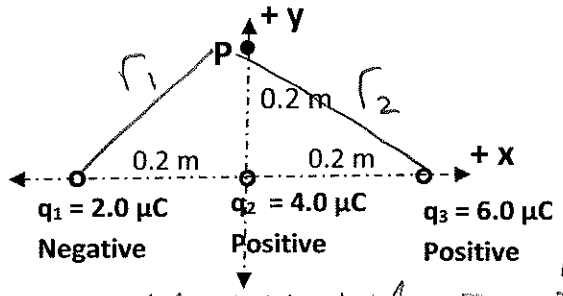
$V_P = 5.4 \times 10^4\,V$

15) Three point charges of 2.0 μC negative, 4.0 μC positive, and 6.0 μC positive are placed along the *x*-axis as shown in the figure below. The charges are equally spaced 0.2 m apart. What is the electrical potential at point P 0.2 m along the + y axis due to these charges?

$r_1 = \sqrt{(0.2m)^2 + (0.2m)^2} = 0.28m$

$r_2 = 0.2\ m$

$r_3 = 0.28m$



$V_P = V_1 + V_2 + V_3 = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3}$

$V_P = (-64{,}300V) + (180{,}000V) + (193{,}000V) = 309{,}000V$

16) A battery has a voltage of 12 V, how much work can one Coulomb of charge do from this battery?

(A) 12 J
B) 6 J
C) 0 J
D) 6 J
E) 144 J

$\Delta V = \frac{W}{q} \quad | \quad W = q\Delta V = 12J$

17) A light bulb is wired to a 10 V battery and the bulb has 2 Coulombs of charge flow throw it, how much energy did the bulb uses?

A) 0 J
B) 10 J
C) 5 J
(D) 20 J
E) 40 J

$W = q\Delta V = 20J$

18) A surface on which all points are at the same potential is referred to as

A) a dielectric surface.
B) a constant electric field surface.
C) an equal voltage surface.
D) a constant electric force surface.
(E) an equipotential surface.

19) An equipotential surface must be

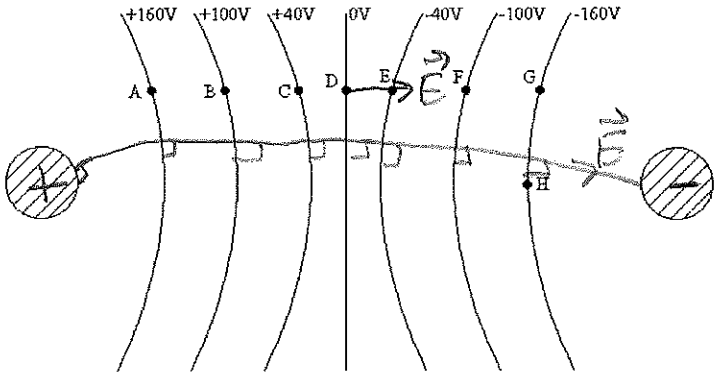
A) parallel to the electric field at any point.
(B) perpendicular to the electric field at any point.
C) equal to the electric field at any point.
D) randomly oriented with respect to the electric field.

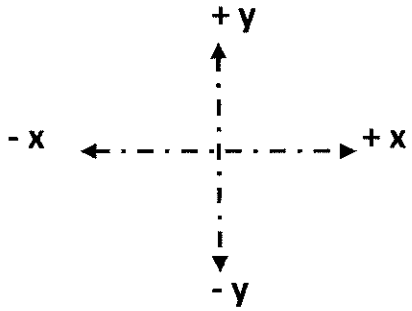
20) The closer the equipotential surfaces are to each other the

$|\vec{E}| = \frac{\Delta V}{\Delta x} = \frac{dV}{dx}$

(A) larger the change in potential and the larger the electric field strength.
B) larger the change in potential and the smaller the electric field strength.
C) smaller the change in potential and the larger the electric field strength.
D) smaller the change in potential and the smaller the electric field strength.

Use the figure below to answer problems 21 - 25





21) The equipotential surfaces between two spherical conductors are shown in the above figure, with the value of the potential marked for each line. What is the potential difference between points A and G?

$V_A = +160V$ or $-160V$ | $\Delta V_{A-G} = V_G - V_A$

$\Delta V_{A-G} = -320V$

22) How much work is required to move a ball with a net positive charge of 4 μC from point G to point A ?

$W = -q\Delta V = (4\times10^{-6}C)(320V) = 1.28\times10^{-3}J$

23) How much work is required to move the same 4 μC charge from point G to point H?

$\Delta V = 0V$ $\quad W = 0J$

24) Which direction is the electric field pointing at point D on the above diagram?

A) There is no electric field at point D
B) It is in the negative *x* direction.
C) It is in the positive *y* direction.
(D) It is in the positive *x* direction.

25) If an electron was placed a point D in the above diagram and released from rest which direction, would it go, towards point A or point G and how fast would it be going when it got there? (Electron: $e^- = 1.6 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg)



Towards A

$D' = V_A - V_B = 160V$

$\Delta PE_E + \Delta KE = 0$

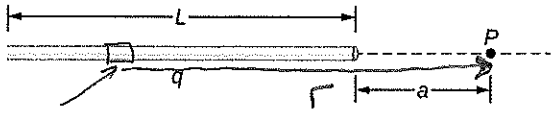
$\Delta PE = -q\Delta V$

$\Delta KE = \frac{1}{2}mv_f^2 - \frcancel{\frac{1}{2}mv_i^2}$

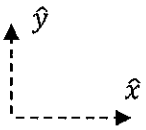
$-q\Delta V + \frac{1}{2}mv_f^2 = 0$

$v_f = \sqrt{\frac{2q\Delta V}{m}} = 7.5\times10^6 m/s$

26) A Thin wire of length L with a uniform linear charge density of λ is shown below, determine the electric potential (V) along the x-axis at point P due to the charge on the wire. Draw and label on the diagram the incremental charge, radius, angle and any other variables needed to solve the problem.



$$V(r) = K\int \frac{dq}{r}$$



$$dq = \lambda dL = \lambda dr$$

$$\int \frac{1}{x}\,dx = \ln(x)$$

$$V(r) = K\int \frac{dq}{r} = K\int_a^{L+a} \frac{\lambda\, dr}{r} = K\lambda\int_a^{L+a} \frac{dr}{r}$$

$$V(r) = K\lambda\left[\ln(r)\right]_a^{L+a} = K\lambda\left[\ln(L+a) - \ln(a)\right]$$

$$V(r) = K\lambda \ln\left(\frac{(L+a)}{a}\right)$$

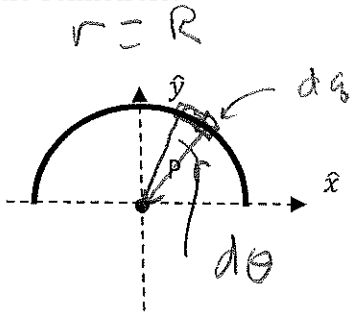
27) Shown below is a semicircular thin wire with radius R and a uniform linear charge density of λ.

A) Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.

B) Determine the electric potential (V) at point P which is at the radius of the semicircle.

$$dq = \lambda dL \rightarrow dL = R\,d\theta$$

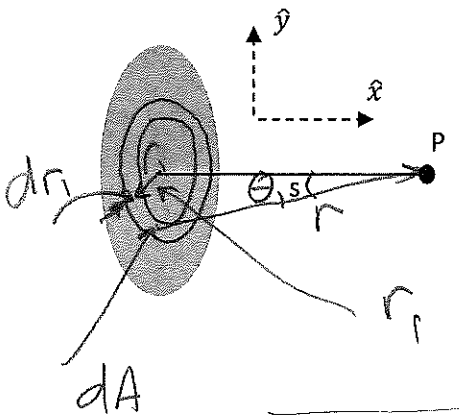
$$dq = \lambda R\,d\theta$$



$$V(r) = K\int \frac{dq}{r} = K\int_0^{\pi} \frac{\lambda R\,d\theta}{R}$$

$$V(r) = K\lambda\int_0^{\pi} d\theta = K\lambda\pi$$

28) A circular disk with a radius R and a uniform surface charge density of σ is shown below, determine the electric potential at point P which is at a distance of s from the center of the disk. Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.



$dA = (2\pi r_1)\,dr_1$ (circumference of dA circle)

$V(r) = K\int \frac{dq}{r}$

$r^2 = s^2 + r_1^2$

$dq = \sigma\, dA$

$dq = \sigma (2\pi r_1)\, dr_1$

$$V(r) = K\int \frac{dq}{r} = K\int_0^R \frac{\sigma(2\pi r_1)\,dr_1}{(s^2+r_1^2)^{1/2}}$$

$$\int \frac{x}{(x^2+a^2)^{1/2}}\,dx = \sqrt{x^2+a^2}$$

$$V(r) = K\sigma 2\pi \int_0^R \frac{r_1\,dr_1}{(s^2+r_1^2)^{1/2}} = K\sigma 2\pi \left[\sqrt{r_1^2+s^2}\right]_0^R$$

$$V(r) = K\sigma 2\pi \left[\sqrt{R^2+s^2} - \sqrt{s^2}\right]$$

$$V(r) = K\sigma 2\pi \left[\sqrt{R^2+s^2} - s\right]$$

Answers

Answers

1) B
2) 0.0025 J
3) 1.37×10^7 J
4) 2.05×10^5 m/s
5) C
6) C
7) B
8) + 9000 V
9) – 9000 V
10) 3×10^{-8} C
11) A
12) – 36,000 V
13) 0 V
14) $+ 5.4 \times 10^4$ V
15) +309,000 V
16) A
17) D
18) E
19) B
20) A
21) - 320 V
22) 1.28×10^{-3} J
23) 0 J
24) D
25) 7.5×10^6 m/s
26)

$$V(r) = k\lambda\left[\frac{(L+a)}{a}\right]$$

27)

$$V(r) = k\lambda\pi$$

28)

$$V(r) = k\sigma 2\pi\left[\sqrt{R^2 + s^2} - s\right]$$